21-242 Recitation 12

Dimension — Q2. Define (V)

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Define $\dim(V)$.

Course-level solution

Definition. If V has a finite basis, the *dimension* $\dim V$ is the length of any basis of V. This is well-defined by Q1: two bases are independent lists each spanning, so each is no longer than the other.
If V has no finite spanning list, one writes $\dim V = \infty$. $\dim\{\vec{0}\} = 0$, the length of the empty basis.